

TOAE209/TOAH209 – Design and Analysis of Algorithms

Week 10: Theoretical Questions

Foreign Trade University

Instructions

Part A contains 16 questions requiring written explanations or short proofs. Part B is a 10-question quiz (true/false and multiple choice). Attempt every question on your own before consulting Part C (Solutions) at the end of this document.

Part A: Theory Questions

- A1.** Define a *flow network* precisely: what are its components (graph, capacities, source, sink), and what simplifying assumptions do we usually make about s and t ?
- A2.** Define an s - t *flow* and the *value* of a flow. State the two conditions (capacity and conservation) precisely, and explain why conservation is required at internal nodes but *not* at s and t .
- A3.** Prove that for any s - t flow, the net flow out of the source equals the net flow into the sink.
- A4.** Define an s - t *cut* (S, T) and its *capacity* $c(S, T)$. Emphasize which edges contribute to the capacity and which do not.
- A5.** Prove the *flow-across-a-cut* identity: for any flow f and any cut (S, T) , $v(f)$ equals the flow on $S \rightarrow T$ edges minus the flow on $T \rightarrow S$ edges.
- A6.** Prove *weak duality*: for any flow f and any cut (S, T) , $v(f) \leq c(S, T)$. What does this immediately imply about $\max_f v(f)$ versus $\min_{(S, T)} c(S, T)$?
- A7.** Explain why a naive greedy “find any s - t path with spare capacity and saturate it” algorithm can fail to find the maximum flow. Give a concrete small example (capacities and a bad path choice) where it gets stuck below the optimum.
- A8.** Define the *residual graph* G_f of a flow f : which forward and backward edges does it contain, and what are their residual capacities? What is the *purpose* of the backward edges?
- A9.** Define an *augmenting path* and its *bottleneck*. Prove that augmenting a flow f along an augmenting path with bottleneck b yields a valid flow of value $v(f) + b$.
- A10.** State the *Max-Flow Min-Cut Theorem*. Then prove the direction that does the real work: if the residual graph G_f has no s - t path, exhibit a cut (S, T) with $c(S, T) = v(f)$, and explain why this proves f is a maximum flow.
- A11.** Explain how, given a maximum flow, you extract a *minimum cut* from it. Why is exactly one BFS in the final residual graph sufficient?

- A12.** State and prove the *Integrality Theorem*: with integer capacities, Ford-Fulkerson terminates and produces an integral maximum flow. Why does this matter for applications such as bipartite matching?
- A13.** State the *Edmonds-Karp* rule (which augmenting path to choose) and its running-time bound. Summarize at a high level why this choice bounds the number of augmentations by $O(VE)$, independent of the capacity values.
- A14.** Describe the reduction from *maximum bipartite matching* to maximum flow: what nodes and edges (with what capacities) do you add? Why does an integral max flow correspond to a matching, and why does its value equal the maximum matching size?
- A15.** Describe the *multiple-sources, multiple-sinks* reduction to a single-source, single-sink max-flow instance (the super-source / super-sink construction). What capacity do you put on the new edges, and why?
- A16.** State *Menger's theorem* (the edge-disjoint-paths form) and explain how it follows from the Max-Flow Min-Cut theorem applied to a unit-capacity network. Why does integrality guarantee that the flow decomposes into edge-disjoint paths?

Part B: Quiz

- B1. (True/False)** In an s - t flow, conservation (flow in = flow out) must hold at every node, including the source and the sink.
- B2. (Multiple choice)** The capacity of an s - t cut (S, T) is the sum of capacities of:
- (a) all edges with at least one endpoint in S
 - (b) edges directed from T to S
 - (c) edges directed from S to T
 - (d) all edges of the graph
- B3. (True/False)** For any flow f and any cut (S, T) , it is always true that $v(f) \leq c(S, T)$.
- B4. (Multiple choice)** A backward residual edge (v, u) in G_f (paired with an original edge (u, v)) exists with residual capacity equal to:
- (a) $c(u, v) - f(u, v)$
 - (b) $f(u, v)$
 - (c) $c(u, v)$
 - (d) always ∞
- B5. (Short answer)** A flow f is a maximum flow if and only if its residual graph G_f has what property?
- B6. (True/False)** If all capacities are integers, then some maximum flow assigns an integer value of flow to every edge.
- B7. (Multiple choice)** Edmonds-Karp chooses, at each step, an augmenting path that is:
- (a) of maximum bottleneck capacity
 - (b) a shortest path (fewest edges) found by BFS
 - (c) found by depth-first search
 - (d) chosen uniformly at random
- B8. (Short answer)** What is the running-time bound of Edmonds-Karp in terms of V (number of nodes) and E (number of edges)?
- B9. (Multiple choice)** To reduce maximum bipartite matching to max flow, the capacity placed on every edge (source $\rightarrow$ left, left $\rightarrow$ right, right $\rightarrow$ sink) is:
- (a) ∞
 - (b) 1
 - (c) the degree of the vertex
 - (d) $|V|$
- B10. (Short answer)** In the multiple-sources/multiple-sinks reduction, what capacity is placed on the edges from the super-source to the original sources (and from the original sinks to the super-sink), and why?

Part C: Solutions

Part A

- A1.** A **flow network** is a directed graph $G = (V, E)$ with a non-negative **capacity** $c(e) \geq 0$ on each edge, a distinguished **source** s , and a distinguished **sink** $t \neq s$. Common simplifying assumptions: no edge enters s and no edge leaves t (so all flow originates at s and terminates at t), and no parallel edges in the same direction. Capacities are usually taken to be integers.
- A2.** An s - t **flow** is a function $f : E \rightarrow \mathbb{R}_{\geq 0}$ with (i) *capacity constraints* $0 \leq f(e) \leq c(e)$ for every edge e , and (ii) *conservation* at every node $v \neq s, t$: total flow into v equals total flow out of v . The **value** is $v(f) = (\text{flow out of } s) - (\text{flow into } s)$. Conservation is *not* imposed at s and t precisely because s is a net *producer* of flow and t a net *consumer*; the whole point is that flow leaves s and arrives at t , so in-/out-flow there are deliberately unbalanced (the imbalance is the value).
- A3.** Sum the conservation equation over all internal nodes $v \neq s, t$. Each edge $e = (u, w)$ with both endpoints internal contributes $+f(e)$ (out of u) and $-f(e)$ (into w), cancelling. After cancellation only terms incident to s or t remain, and rearranging yields (net out of s) = (net into t). (Formally this is the special case $S = \{s\} \cup (\text{internal nodes})$ of the cut identity in A5, or directly $S = V \setminus \{t\}$.)
- A4.** An s - t **cut** is a partition (S, T) of V with $s \in S$, $t \in T$. Its **capacity** is $c(S, T) = \sum_{e=(u,v): u \in S, v \in T} c(e)$ – the total capacity of edges directed *from* S *to* T only. Edges directed from T to S , and edges internal to S or to T , do *not* contribute.
- A5.** Start from $v(f) = \sum_{e \text{ out of } s} f(e) - \sum_{e \text{ into } s} f(e)$. For each $v \in S$ with $v \neq s$, add the (zero) quantity $\sum_{e \text{ out of } v} f(e) - \sum_{e \text{ into } v} f(e)$; the total is unchanged. Regroup the resulting sum by edges: an edge with both ends in S contributes $+f(e)$ and $-f(e)$ (cancels); an $S \rightarrow T$ edge contributes $+f(e)$; a $T \rightarrow S$ edge contributes $-f(e)$. Hence $v(f) = \sum_{e:S \rightarrow T} f(e) - \sum_{e:T \rightarrow S} f(e)$.
- A6.** By A5, $v(f) = \sum_{e:S \rightarrow T} f(e) - \sum_{e:T \rightarrow S} f(e)$. Since $f(e) \geq 0$, the subtracted term is ≥ 0 , so $v(f) \leq \sum_{e:S \rightarrow T} f(e)$. Since $f(e) \leq c(e)$, this is $\leq \sum_{e:S \rightarrow T} c(e) = c(S, T)$. Therefore $v(f) \leq c(S, T)$. Taking the maximum over f and the minimum over cuts: $\max_f v(f) \leq \min_{(S,T)} c(S, T)$ (the max-flow is at most the min-cut).
- A7.** Without the ability to reroute, an early path can saturate an edge that a better solution needed differently. Example: capacities all 1 on edges $s \rightarrow a$, $s \rightarrow b$, $a \rightarrow b$, $a \rightarrow t$, $b \rightarrow t$. Greedy might pick $s \rightarrow a \rightarrow b \rightarrow t$, pushing 1 unit and saturating $a \rightarrow b$; now $a \rightarrow t$ and $b \rightarrow t$ each still have capacity, but a can no longer receive more from s except via $s \rightarrow a$ (already used), so the naive method stops at value 1. The true maximum is 2 ($s \rightarrow a \rightarrow t$ and $s \rightarrow b \rightarrow t$), reachable only by “undoing” the use of $a \rightarrow b$ – exactly what residual backward edges allow.
- A8.** The **residual graph** G_f has the same nodes and: for each original edge (u, v) with $f(u, v) < c(u, v)$, a *forward* residual edge (u, v) with residual capacity $c(u, v) - f(u, v)$ (spare we can still push); and for each original edge (u, v) with $f(u, v) > 0$, a *backward* residual edge (v, u) with residual capacity $f(u, v)$ (the amount we could cancel). The backward edges are what let the algorithm *retract* poorly-placed flow, escaping the dead-ends that defeat naive greedy.

- A9.** An **augmenting path** is a simple s - t path in G_f ; its **bottleneck** b is the minimum residual capacity along it. *Proof that augmenting gives a valid flow of value $v(f) + b$:* On each forward edge of P we add $b \leq c(e) - f(e)$, so $f' \leq c$; on each backward edge we subtract $b \leq f(e)$, so $f' \geq 0$ – capacity constraints hold. At each internal node of P , the path uses exactly one incoming and one outgoing residual edge, and both change the node’s net balance by $+b$ and $-b$ (or are arranged so the change cancels), preserving conservation; nodes off P are untouched. The first edge of P leaves s (forward, since nothing enters s), so the net out of s increases by exactly b , giving $v(f') = v(f) + b$.
- A10. Max-Flow Min-Cut Theorem:** the maximum value of an s - t flow equals the minimum capacity of an s - t cut; equivalently, f is maximum iff G_f has no augmenting path. *Proof of the key direction:* suppose G_f has no s - t path. Let S be the set of nodes reachable from s in G_f and $T = V \setminus S$; then $s \in S$, $t \in T$. Every original edge (u, v) with $u \in S$, $v \in T$ must be *saturated* ($f = c$): otherwise a forward residual edge (u, v) would put v in S . Every original edge (v, u) with $v \in T$, $u \in S$ must carry *zero* flow: otherwise a backward residual edge (u, v) would put v in S . By A5, $v(f) = \sum_{e:S \rightarrow T} f(e) - \sum_{e:T \rightarrow S} f(e) = \sum_{e:S \rightarrow T} c(e) - 0 = c(S, T)$. By weak duality every flow is $\leq c(S, T) = v(f)$, so f is maximum (and (S, T) is a minimum cut).
- A11.** After computing a maximum flow f , run a single BFS (or DFS) from s in the final residual graph G_f , following only edges with positive residual capacity. Let S be the set of nodes reached; then $(S, V \setminus S)$ is a minimum cut (this is exactly the construction in A10). One traversal suffices because “reachable from s in G_f ” is a single connectivity query, and the proof in A10 shows that the resulting S already gives a cut whose capacity equals $v(f)$ – which weak duality certifies as minimum.
- A12. Integrality Theorem:** with integer capacities, Ford-Fulkerson terminates and yields an integral maximum flow. *Proof:* the initial flow $f \equiv 0$ is integral. If f is integral, all residual capacities $c(e) - f(e)$ and $f(e)$ are integers, so the bottleneck b of any augmenting path is a positive integer; augmenting preserves integrality. Each augmentation raises $v(f)$ by $b \geq 1$, and $v(f)$ is bounded by the finite source-cut capacity, so only finitely many augmentations occur – termination with an integral flow. *Why it matters:* reductions like bipartite matching use unit capacities, and integrality guarantees the resulting flow is 0/1 on every edge, so it can be read directly as a yes/no selection (a matching), with no fractional values to round.
- A13. Edmonds-Karp** always augments along a *shortest* augmenting path (fewest edges), found by BFS; this gives an $O(VE^2)$ running time. *Why $O(VE)$ augmentations:* under shortest-path augmentation, BFS distances from s are non-decreasing over time, and each edge can be the saturated “bottleneck” only $O(V)$ times (between two saturations of an edge (u, v) , the distance to u must increase by at least 2, and distances range over $\{0, \dots, V-1\}$). With $O(VE)$ saturations and at least one saturation per augmentation, there are $O(VE)$ augmentations; each BFS costs $O(E)$, for $O(VE^2)$ total – crucially independent of capacity magnitudes.
- A14.** Build a flow network: a super-source s with a capacity-1 edge to each left-vertex; a capacity-1 edge (left u , right v) for each allowed pair; and a capacity-1 edge from each right-vertex to a super-sink t . By integrality the max flow is 0/1 on every edge; the saturated left→right edges form a matching (the capacity-1 edges at each left and right vertex force at most one matched partner per vertex – the conservation/capacity constraints encode “each vertex used at most once”). Each unit of flow corresponds to one matched pair, so the max-flow value equals the maximum matching size.

- A15.** Add a **super-source** S^* with an edge $S^* \rightarrow s_i$ to every original source s_i , and a **super-sink** T^* with an edge $t_j \rightarrow T^*$ from every original sink t_j . Give these new edges capacity $+\infty$ (or any value $\geq$ the total possible flow), so they never constrain the solution – they merely *collect* flow from all sources and *deliver* it to all sinks. A max flow from S^* to T^* then equals the maximum total flow the original sources can jointly deliver to the original sinks.
- A16. Menger’s theorem (edge form):** the maximum number of pairwise *edge-disjoint* s - t paths equals the minimum number of edges whose removal disconnects t from s (the minimum s - t edge cut). *From max-flow min-cut:* give every edge capacity 1. Then the min-cut capacity is exactly the minimum number of edges separating s from t , and max-flow min-cut equates it with the max-flow value. By integrality the max flow is 0/1 on each edge; the edges carrying flow form a subgraph in which every internal node has equal in/out flow, so it *decomposes* into exactly $v(f)$ paths from s to t , and since each edge carries at most 1 unit these paths are edge-disjoint. Hence (number of edge-disjoint paths) = $v(f)$ = (min edge cut).

Part B

- B1. False.** Conservation is required only at internal nodes; s and t are deliberately exempt, since s produces and t consumes the flow (their imbalance is the flow value).
- B2. (c) edges directed from S to T .** Only forward-crossing edges count; edges from T to S are free.
- B3. True.** This is weak duality (A6): every cut upper-bounds every flow.
- B4. (b) $f(u, v)$.** A backward residual edge (v, u) has residual capacity equal to the current flow on (u, v) – the amount that can be cancelled.
- B5. G_f has no augmenting path** (no s - t path in the residual graph). Equivalently, t is unreachable from s in G_f .
- B6. True.** This is the Integrality Theorem (A12): integer capacities yield an integral maximum flow.
- B7. (b) a shortest path (fewest edges) found by BFS.** That is precisely the Edmonds-Karp rule.
- B8. $O(VE^2) - O(VE)$ augmentations, each with an $O(E)$ BFS.**
- B9. (b) 1.** Unit capacities on all three edge types ensure each vertex is matched at most once and (with integrality) the flow reads off as a matching.
- B10.** Capacity $+\infty$ (or any value at least the total achievable flow). These edges should never be the bottleneck; they exist only to merge all sources into one super-source and all sinks into one super-sink, so the true constraints come from the original edges.